Euler's analytic proof.

UFT in $\mathbb{Z}$: $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ uniquely up to order.

Look at

$$\infty = \sum_{n=1}^{\infty} \frac{1}{n} \overset{?}{=} \prod_{p} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3} + \cdots\right)$$

$\frac{1}{1-\frac{1}{p}} = (1-p^{-1})^{-1}$

p ↑ prime

E.g. $n = 300 = 2^2 \cdot 3 \cdot 5^2$.

$$\frac{1}{n} = \frac{1}{2^2} \cdot \frac{1}{3} \cdot \frac{1}{5^2} \cdot 1$$

$$\left(1 + \frac{1}{2} + \frac{1}{2^2} + \cdots\right)\left(1 + \frac{1}{3} + \cdots\right)\left(1 + \frac{1}{5} + \frac{1}{5^2} + \cdots\right)\left(1 + \frac{1}{7} + \cdots\right) \cdots$$

Conclusion:

$$\sum_{n=1}^{\infty} \frac{1}{n} = \prod_{p} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3} + \cdots\right).$$

Thm (Euler) The series

$$\sum_p \frac{1}{p}$$

diverges. In particular, there are infinitely many primes.

Two inequalities:

1. $x < \ln(1-x)^{-1} < x + x^2$ for $0 < x \leq \frac{1}{2}$

Recall: $\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots$, $|x| < 1$.

Replacing x by $-x$ and multiplying by -1:

$$\ln(1-x)^{-1} = -\ln(1-x) = x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \frac{1}{4}x^4 + \cdots$$

When $0 < x < \frac{1}{2}$, $\ln(1-x)^{-1} > x$ and

$$\ln(1-x)^{-1} = x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \frac{1}{4}x^4 + \cdots$$

$$= x + x^2\left(\frac{1}{2} + \frac{1}{3}x + \frac{1}{4}x^2 + \cdots\right)$$

$0 < x \le \frac{1}{2}$

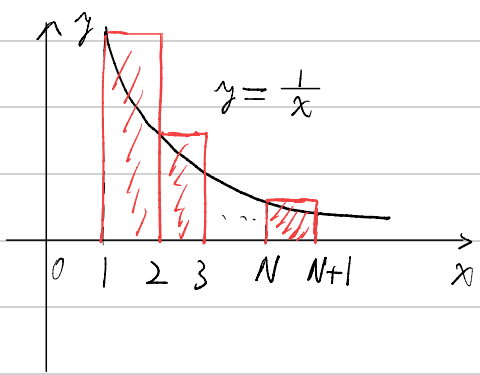
$$< x + x^2 \cdot \frac{1}{2}\left(1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \cdots\right)$$

$$\frac{1}{1-\frac{1}{2}} = 2$$

$$= x + x^2.$$

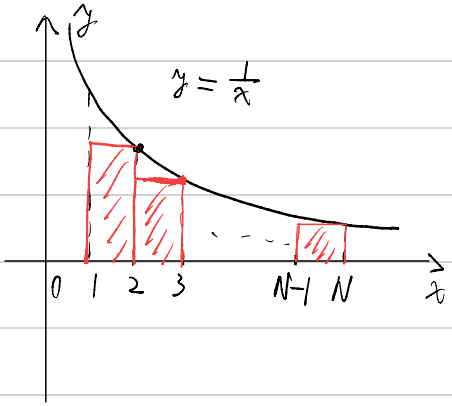
2. $\ln N < \sum_{n=1}^{N} \frac{1}{n} < 1 + \ln N$ for all integers $N > 1$.

Think about the integral test:



$$\sum_{n=1}^{N} \frac{1}{n} > \int_1^N \frac{1}{x}\,dx.$$

$$= \ln N.$$



$$\boxed{\sum_{n=2}^{N} \frac{1}{n}} < \int_1^N \frac{1}{x}\,dx$$

$$= \ln N.$$

$$\Rightarrow \sum_{n=1}^{N} \frac{1}{n} < 1 + \ln N.$$

Now we return to Euler's theorem.

Look at

$n \le N$ has every prime factor $\le N$ ↓

$$\ln N < \sum_{n=1}^{N} \frac{1}{n} \le \sum_{n=1}^{\infty} \frac{1}{n} = \prod_{p \le N} \underbrace{\left(1 + \frac{1}{p} + \frac{1}{p^2} + \cdots\right)}_{(1-p^{-1})^{-1}}$$

$p \mid n \Rightarrow p \le N$

Taking $\ln$:

$$\ln(\ln N) < \ln \prod_{p \le N} (1 - p^{-1})^{-1} = \sum_{p \le N} \ln(1 - p^{-1})^{-1}.$$

Since $x = p^{-1} \leq \frac{1}{2}$,

$$\ln(1-p^{-1})^{-1} < p^{-1} + p^{-2},$$

we have

$$\ln(\ln N) < \sum_{p \leq N} \ln(1-p^{-1})^{-1}$$

$$< \sum_{p \leq N} (p^{-1} + p^{-2})$$

$$= \sum_{p \leq N} \frac{1}{p} + \sum_{p \leq N} \frac{1}{p^2}$$

$$\leq C = \sum_{p} \frac{1}{p^2} < \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

(integral test.)

So $\sum_{p \leq N} \frac{1}{p} > \ln(\ln N) - C$, where $N \geq 3$.

Hence, letting $N \to \infty$, $\sum_{p} \frac{1}{p} = \infty$.

Define

$$\pi(N) = \text{number of primes} \leq N.$$

We already know

$$\lim_{N \to \infty} \pi(N) = \infty.$$

Next time, we will study the growth of $\pi(N)$.